A whole lot of reductions

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1 preliminaries

Definition 1. *A **tour** in a graph $G = (V, E)$ is a sequence of vertices and edges adjacent to them in the graph that spans every vertex. We usually refer to the multiset of edges defined by a tour as T. The **cost** of a tour T is the sum of the weights of the edges in T, denoted as $c(T) = \sum_{e \in T} c(e)$.*

This definition is equivalent to saying that a tour is a Eulerian multigraph that spans every vertex of G.

Definition 2 (Metric TSP)**.** *Let $G = (V, E)$ be a connected undirected graph. The **Metric Traveling Salesperson Problem** (Metric TSP) is defined as follows:*

- ***Instance:*** *A complete graph $G = (V, E)$ which is connected by definition, and a cost function $c : E \to \mathbb{R}_{\geq 0}$ that satisfies the triangle inequality:*

$$c(u, v) \leq c(u, w) + c(w, v) \quad \text{for all } u, v, w \in V. \tag{1}$$

- ***Objective:*** *Find a closed tour C in G such that the total weight $c(C) = \sum_{e \in C} c(e)$ is minimized.*

Definition 3 (Quasi-Tour)**.** *A **quasi-tour** T_0 in a graph $G = (V, E)$ is a subset of edges in T removing edges from a finite set of closed tours. Each connected component of this multigraph is either a Eulerian multigraph or an isolated vertex.*

A given multiset of edges T_0 is a quasi-tour if and only if it is Eulerian.

2 Starting point

Definition 4 (MAX-E3-LIN-2)**.** *The **MAX-E3-LIN-2** problem is defined as follows:*

- ***Instance:*** *A set of n boolean variables $x_1, x_2, \ldots, x_n$ and a collection of m linear equations over $\mathbb{F}_2$, where each equation involves exactly three variables. Each equation is of the form:*

$$x_{i_1} + x_{i_2} + x_{i_3} = b_j, \tag{2}$$

  where $b_j \in \{0, 1\}$ and i_1, i_2, i_3 are distinct indices from $\{1, 2, \ldots, n\}$.

- ***Objective:*** *Find an assignment to the variables that maximizes the number of satisfied equations.*